

Chapter 1.

#1.1.1. Let $f = x^2y$ and $g = y \cdot \sin z$ be functions on $\mathbb{R}^3$. Express the followings:

a). $f g^2$

sol). $f g^2 = x^2y \cdot y \sin^2 z = (xy)^2 \sin^2 z$.

b) $\frac{\partial f}{\partial x} g + \frac{\partial g}{\partial y} f = 2xy \cdot y \sin z + \sin z \cdot x^2y = (\sin z) \cdot xy(x + 2y)$

c) $\frac{\partial^2 (fg)}{\partial y \partial z} = \frac{\partial}{\partial y} (x^2y^2 \cos z) = 2x^2y \cos z$.

d) $\frac{\partial}{\partial y} \sin f = \cos(x^2y) \cdot x^2$.

#1.2.1. Let $V = (-2, 1, -1)$ and $W = (0, 1, 3)$.

a). Given $P \in \mathbb{R}^3$, express the tangent vector $3V_P - 2W_P$ as a linear combination of $U_1(P)$, $U_2(P)$, $U_3(P)$.

sol). observe $V_P = (-2, 1, -1)_P = -2 \cdot (1, 0, 0)_P + 1 \cdot (0, 1, 0)_P + (-1) \cdot (0, 0, 1)_P$
 $W_P = (0, 1, 3)_P = 0 \cdot (1, 0, 0)_P + 1 \cdot (0, 1, 0)_P + 3 \cdot (0, 0, 1)_P$,

so $3V_P - 2W_P = -6 \cdot (1, 0, 0)_P + 1 \cdot (0, 1, 0)_P + (-7) \cdot (0, 0, 1)_P$.

#1.2.2. Let $V = xU_1 + yU_2$ and $W = 2x^2U_2 - U_3$.

Compute the vector field $W - xV$, and find its value at the point $P = (-1, 0, 2)$.

Proof. $W - xV = 2x^2U_2 - U_3 - (x^2U_1 + xyU_2)$
 $= -x^2U_1 + (2x^2 - xy)U_2 - U_3$.

Now, $(W - xV)(-1, 0, 2) = (1, 2, -1)_P$.

Example 3.3, sec 2.3.

Consider the unit-speed helix

$$\beta(s) = \left(a \cdot \cos \frac{s}{c}, a \cdot \sin \frac{s}{c}, \frac{b}{c} \cdot s \right) \text{ where } a > 0, c = \sqrt{a^2 + b^2}.$$

Figure out the Frenet Frame.

$$T(s) = \beta'(s) = \left(-\frac{a}{c} \sin \frac{s}{c}, \frac{a}{c} \cos \frac{s}{c}, \frac{b}{c} \right)$$

$$\hookrightarrow T'(s) = \beta''(s) = \left(-\frac{a}{c^2} \cos \frac{s}{c}, -\frac{a}{c^2} \sin \frac{s}{c}, 0 \right) = -\frac{a}{c^2} \cdot \left(\cos \frac{s}{c}, \sin \frac{s}{c}, 0 \right)$$

$$\hookrightarrow K(s) = \|T'(s)\| = \frac{a}{c^2} = \frac{a}{a^2 + b^2} > 0.$$

$$\hookrightarrow N(s) = \frac{a^2 + b^2}{a} \cdot T'(s) = \left(-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0 \right)$$

$$\hookrightarrow B(s) = \left(\frac{b}{c} \cdot \sin \frac{s}{c}, -\frac{b}{c} \cdot \cos \frac{s}{c}, \frac{a}{c} \right)$$

$$\hookrightarrow B'(s) = \left(\frac{b}{c^2} \cos \frac{s}{c}, \frac{b}{c^2} \sin \frac{s}{c}, 0 \right)$$

$$\hookrightarrow \tau(s) = \frac{b}{c^2} = \frac{b}{a^2 + b^2}.$$



Geometrically, if $b=0$, then the Curvature $K=\frac{1}{a}$, and the torsion is Zero.

Sec 1.7 mapping.

#124, let $F(u, v) = (u^2 - v^2, 2uv)$.

1). Find all points P s.t.

a) $F(P) = (0, 0)$

sol). $\begin{cases} u^2 - v^2 = 0 \\ 2uv = 0 \end{cases}$ $uv = 0$ implies either $u = 0$ or $v = 0$, so $u = v = 0$, by $u^2 = v^2$.

c) $F(P) = P$

sol). $(u^2 - v^2, 2uv) = (u, v)$

$2uv = v \Rightarrow v(2u - 1) = 0$. So either $v = 0$ or $u = \frac{1}{2}$.

If $v = 0$, $u^2 - v^2 = u$ so $u = 0$ or 1

If $u = \frac{1}{2}$, then $v^2 = \frac{1}{4}$, so $v = \pm \frac{1}{2}$,

So the tuples $(0, 0), (0, 1), (\frac{1}{2}, \pm \frac{1}{2})$ are the answer.

#3. Let $V = (V_1, V_2)$ and $P = (P_1, P_2)$. Express $F_*(V_P)$ directly.

sol). $F_*(V_P) = (V_P[u^2 - v^2], V_P[2uv])$

for $f_1(u, v) = u^2 - v^2$,

$$\begin{aligned} V_P[f_1] &= \lim_{t \rightarrow 0} \frac{f_1(P_1 + tV_1, P_2 + tV_2) - f_1(P_1, P_2)}{t} \\ &= \lim_{t \rightarrow 0} \frac{(P_1 + tV_1)^2 - (P_2 + tV_2)^2 - (P_1^2 - P_2^2)}{t} \\ &= \lim_{t \rightarrow 0} \frac{2tP_1V_1 - 2tP_2V_2 + t^2(V_1^2 + V_2^2)}{t} \\ &= 2P_1V_1 - 2P_2V_2 \end{aligned}$$

(Or, by the lemma, $V_1 \cdot 2u|_{u=P_1} + V_2 \cdot (-2v|_{v=P_2}) = 2P_1V_1 - 2P_2V_2$.

And similarly $V_P[f_2] = V_1 \cdot 2P_2 + V_2 \cdot 2P_1$. Consequently,

$$F_*(V_P) = (2P_1V_1 - 2P_2V_2, 2P_2V_1 + 2P_1V_2)$$

#5. If $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, then $F_*(V_p) = F(V)_{F(p)}$
 Proof. By the Theorem,

$$F_*(V_p) = (V_p[f_1], \dots, V_p[f_m])_{F(p)}$$

And, since F is linear, each $f_i: \mathbb{R}^n \rightarrow \mathbb{R}$ also linear functional.

Therefore,
$$V_p[f_i] = \lim_{t \rightarrow 0} \frac{f_i(p + tV) - f_i(p)}{t} = \lim_{t \rightarrow 0} \frac{t \cdot f_i(V)}{t} = f_i(V). \quad (1 \leq i \leq m)$$

So
$$F_*(V_p) = (f_1(V), \dots, f_m(V))_{F(p)} = F(V)_{F(p)}.$$

#7. If $V_p \in T_p \mathbb{R}^n$ and $\vartheta: \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable,
 then for a mapping $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $F_*(V_p)[\vartheta] = V_p[\vartheta(F)]$.

That is, a mapping preserves directional derivatives.

Proof. By the Theorem, for $F = (f_1, \dots, f_m)$,

$$F_*(V_p) = (V_p[f_1], \dots, V_p[f_m])_{F(p)}, \quad \text{so therefore}$$

$$F_*(V_p)[\vartheta] = \sum_{i=1}^m V_p[f_i] \cdot \frac{\partial \vartheta}{\partial x_i}(F(p)).$$

And, by the chain rule,

$$V_p[\vartheta(F)] = \sum_{i=1}^m V_i \frac{\partial \vartheta \circ F}{\partial x_i}(p) = \sum_{i=1}^m V_i \cdot \left[\frac{\partial \vartheta}{\partial F} \cdot \frac{\partial F}{\partial x_i} \right](p)$$

Exercise 4.1.

#4.1.1. Prove that the followings are Not surfaces. (Formal proof is not required.)

1. Cone $z^2 = x^2 + y^2$.

2. Closed disk: $x^2 + y^2 \leq 1, z = 0$.

3. Folded plane: $xy = 0, x \geq 0, y \geq 0$.



2. Describe a surface $M: ax + by + cz = d$ (a, b, c not all zero) by vector equation.

Sol). $M = \{(x, y, z) \in \mathbb{R}^3 \mid (a, b, c) \cdot (x, y, z) - d = 0\}$

$$= \{(x, y, z) \in \mathbb{R}^3 \mid$$

* 4.1.4.

Determine whether the following is a patch or not.

a). $X(u, v) = (u, uv, v)$

b). $X(u, v) = (u^2, u^3, v)$

c). $X(u, v) = (u, u^2, v+v^3)$

d). $X(u, v) = (\cos 2\pi u, \sin 2\pi u, v)$

Proof. a) is clearly 1-1, because $(u_1, u_1 v_1, v_1) = (u_2, u_2 v_2, v_2) \Rightarrow u_1 = u_2$ and $v_1 = v_2$.

To show the regularity, calculate the Jacobian:

$$DX = \begin{bmatrix} 1 & 0 \\ v & u \\ 0 & 1 \end{bmatrix}$$

clearly it has rank 2 for all $(u, v) \in \mathbb{R}^2$, so it is regular

b) is similarly 1-1, since $u_1^3 = u_2^3 \Rightarrow u_1 = u_2$. And, again,

$$DX = \begin{bmatrix} 2u & 0 \\ 3u^2 & 0 \\ 0 & 1 \end{bmatrix} \quad v(v^2+1)$$

it is has rank 1 when $u=0$, so it is Not regular.

c) is also 1-1, because $x \mapsto x+x^3$ is 1-1. And,

$$DX = \begin{bmatrix} 1 & 0 \\ 2u & 0 \\ 0 & 1+3v^2 \end{bmatrix}$$

is always rank 2, so regular.

d) is Not 1-1 on $\mathbb{R}^2$, but 1-1 on $(-1, 1) \times \mathbb{R}$. Moreover,

$$DX = \begin{bmatrix} -2\pi \sin 2\pi u & 0 \\ 2\pi \cos 2\pi u & 0 \\ 0 & 1 \end{bmatrix} \text{ is always rank 2}$$

#4, 1, 5. a).

Prove that $M = \{(x, y, z) \in \mathbb{R}^3 \mid (x^2 + y^2)^2 + 3z^2 = 1\}$ is a surface.

Proof. Observe that: for $(x, y, z) \in \mathbb{R}^3$,

$$(x^2 + y^2)^2 = 1 - 3z^2 \text{ has solution only } -\frac{1}{\sqrt{3}} \leq z \leq \frac{1}{\sqrt{3}}$$

Construct four patches: the upper, the below, and the middle.

$$X_1: D_1 \rightarrow \mathbb{R}^3: (u, v) \mapsto \left(u, v, \frac{1}{\sqrt{3}} \cdot \sqrt{1 - (u^2 + v^2)^2}\right)$$

where $D_1 = \{(u, v) \mid u^2 + v^2 < 1\}$.

$$X_2: D_1 \rightarrow \mathbb{R}^3: (u, v) \mapsto \left(u, v, -\frac{1}{\sqrt{3}} \sqrt{1 - (u^2 + v^2)^2}\right)$$

The X_1 and X_2 are Monge patch. And, to cover the middle circle,

$$X_3: D_2 \rightarrow \mathbb{R}^3: (u, v) \mapsto \left(\sqrt[4]{1 - 3v^2} \cdot \cos u, \sqrt[4]{1 - 3v^2} \cdot \sin u, v\right).$$

where $D_2 = (-\pi, \pi) \times \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

and $X_4 = X_3(u + \pi, v + \pi)$

Then, X_1, X_2, X_3, X_4 cover M .